

Chapter 3.

3.1. Since $-X(t+s) - (-X(s))$

$$= -X(t+s) + X(s)$$

$$= -(X(t+s) - X(s))$$

is the negative of a normal random variable with mean μt and variance $\sigma^2 t$, and so $-X(t)$, $t \geq 0$ is a Brownian motion with drift parameter $-\mu$ and variance parameter σ^2 .

3.2. $\mu = 3$, $\sigma^2 = 9$.

(a) $E[X(2)] = \mu \cdot 2 + X(0) = 16$

(b) $\text{Var}(X(2)) = \sigma^2 \cdot 2 = 9 \cdot 2 = 18$

(c) $P(X(2) > 20) = P(Z > \frac{20-16}{\sqrt{18}})$

$$\approx P(Z > 0.9428)$$

$$\approx 0.173$$

(d) $E[X(0.5)] = \mu \cdot 0.5 + 10 = 11.5$

$$\text{Var}(X(0.5)) = \sigma^2 \cdot 0.5 = 4.5$$

$$P(X(0.5) > 10) = P(Z > \frac{10-11.5}{\sqrt{4.5}})$$

$$\approx P(Z > -0.7071)$$

$$= P(Z < 0.7071)$$

$$\approx 0.7602$$

3.3. $\Delta = 0.1$, $P = \frac{1}{2}(1 + \frac{\mu}{\sigma}\sqrt{\Delta}) \approx 0.6581$
 $X_i = \begin{cases} 1 & \text{the change at time } i\Delta \text{ is an increase} \\ -1 & \text{otherwise.} \end{cases}$

$$E[X_i] = 1 \cdot P + (-1)(1-P) = 2P - 1 = \frac{\mu}{\sigma}\sqrt{\Delta}$$

$$= \frac{3}{3} \cdot \sqrt{0.1} = \sqrt{0.1} \approx 0.3162$$

$$\text{Var}(X_i) = E[X_i^2] - E[X_i]^2$$

$$= 1^2 \cdot P + (-1)^2 \cdot (1-P) - (2P-1)^2$$

$$= 1 - \frac{\mu^2 \Delta}{\sigma^2}$$

$$= 0.9$$

thus

(a) $E[X(1)] = E[\sum_{i=1}^{10} \sigma\sqrt{\Delta} X_i] + 10$

$$= \sigma\sqrt{\Delta} \sum_{i=1}^{10} E[X_i] + 10$$

$$= 3 \cdot \sqrt{0.1} \cdot 10 \cdot \sqrt{0.1} + 10$$

$$= 13$$

(b) $\text{Var}[X(1)] = \text{Var}(\sigma\sqrt{\Delta} \sum_{i=1}^{10} X_i)$

$$= (\sigma\sqrt{\Delta})^2 \sum_{i=1}^{10} \text{Var}(X_i)$$

$$= \sigma^2 \cdot \Delta \cdot 10 \cdot 0.9$$

$$= 8.1$$

(c). By time 0.5, there are totally 5 changes. 0.5/0.1 = 5
 To make $X(0.5) > 10$, we must have more positive changes than negative changes. Thus

$P(X(0.5) > 10) = \binom{5}{0} (0.6581)^5 + \binom{5}{1} (0.6581)^4 (1-0.6581)$

$$+ \binom{5}{2} (0.6581)^3 (1-0.6581)^2$$

$$\approx 0.7773$$

3.4. (a) $P(S(1) > S(0)) = P(\frac{S(1)}{S(0)} > 1)$

$$= P(\log(\frac{S(1)}{S(0)}) > 0)$$

$$= P(X(1) > 0)$$

$$= P(Z > \frac{0 - \mu \cdot 1}{\sqrt{\sigma^2 \cdot 1}})$$

$$= P(Z > \frac{-0.1}{\sqrt{0.2^2}})$$

$$= P(Z > -\frac{1}{2})$$

$$= P(Z < \frac{1}{2}) = \Phi(\frac{1}{2}) \approx 0.6915$$

(b) $P(S(2) > S(1) > S(0)) = P(S(2) > S(1)) \cdot P(S(1) > S(0))$

$$= 0.6915 \cdot 0.6915$$

$$\approx 0.4781$$

(c) $P(S(3) < S(1)) = P(\frac{S(3)}{S(1)} < 1)$

$$= P(\log(\frac{S(3)}{S(1)}) < 0)$$

$$= P(X(2) < 0) = P(Z < \frac{0 - \mu \cdot 2}{\sqrt{\sigma^2 \cdot 2}})$$

$$= P(Z < \frac{-0.2}{0.2\sqrt{2}}) = P(Z < -\frac{\sqrt{2}}{2})$$

$$= 1 - P(Z > \frac{\sqrt{2}}{2}) = 1 - \Phi(\frac{\sqrt{2}}{2})$$

$$\approx 0.2389$$

$$\Rightarrow P(S(3) < S(1) > S(0)) = P(S(1) > S(0)) \cdot P(S(3) < S(1))$$

$$\approx 0.6915 \cdot 0.2389$$

$$\approx 0.1652$$

3.6. $S'(t) = se^{X(t)}$ where $X(t)$ is a normal random variable with mean μt and variance $\sigma^2 t$. We know

$$E[e^{X(t)}] = e^{E[X(t)] + \frac{1}{2} \text{Var}(X(t))}$$

(see lecture notes)

Thus

$$\text{Var}(S'(t)) = \text{Var}(se^{X(t)}) = s^2 \text{Var}(e^{X(t)})$$

$$= s^2 (E[e^{2X(t)}] - E[e^{X(t)}]^2)$$

$$= s^2 (E[e^{2\mu t + 2\sigma^2 t}] - E[e^{\mu t + \frac{1}{2}\sigma^2 t}]^2)$$

$$= s^2 (e^{2\mu t + 2\sigma^2 t} - e^{2\mu t + \sigma^2 t})$$

$$= s^2 e^{2\mu t + \sigma^2 t} (e^{\sigma^2 t} - 1)$$

3.7. $\lim_{t \rightarrow \infty} \bar{\Phi}\left(\frac{y - \mu t}{\sigma \sqrt{t}}\right) = 1 - \lim_{t \rightarrow \infty} \Phi\left(\frac{y - \mu t}{\sigma \sqrt{t}}\right) = \begin{cases} 0 & \text{if } \mu < 0 \\ 1 & \text{if } \mu \geq 0 \end{cases}$

$\lim_{t \rightarrow \infty} \bar{\Phi}\left(\frac{y + \mu t}{\sigma \sqrt{t}}\right) = 1 - \lim_{t \rightarrow \infty} \Phi\left(\frac{y + \mu t}{\sigma \sqrt{t}}\right) = \begin{cases} 1 & \text{if } \mu < 0 \\ 0 & \text{if } \mu \geq 0 \end{cases}$

thus by the formula for $P(T_y < t)$, we have

$$P(T_y < +\infty) = \lim_{t \rightarrow \infty} P(T_y < t)$$

$$= \lim_{t \rightarrow \infty} P(M(t) \geq y)$$

$$= \lim_{t \rightarrow \infty} e^{2y\mu/\sigma^2} \bar{\Phi}\left(\frac{y - \mu t}{\sigma \sqrt{t}}\right) + \bar{\Phi}\left(\frac{y + \mu t}{\sigma \sqrt{t}}\right)$$

$$= \begin{cases} e^{2y\mu/\sigma^2} & \text{if } \mu < 0 \\ 1 & \text{if } \mu \geq 0 \end{cases}$$

3.8. $S'(v) = se^{X(v)}$ ($s > 0$)

where $X(v)$ is a Brownian motion with drift parameter μ and volatility parameter σ , and $X(0) = 0$

$$P(\max_{0 \leq v \leq t} S'(v) \geq y)$$

$$= P(\max_{0 \leq v \leq t} se^{X(v)} \geq y)$$

$$= P(\max_{0 \leq v \leq t} e^{X(v)} \geq \frac{y}{s})$$

clearly, if $y \leq 0$, then

$$P(\max_{0 \leq v \leq t} S'(v) \geq y) = 1$$

Assume $y > 0$, then

$$P(\max_{0 \leq v \leq t} S'(v) \geq y) = P(\max_{0 \leq v \leq t} X(v) \geq \log(\frac{y}{s}))$$

$$= P(M(t) \geq \log(\frac{y}{s}))$$

$$= e^{2\log(\frac{y}{s})\mu/\sigma^2} \bar{\Phi}\left(\frac{\log(\frac{y}{s}) + \mu t}{\sigma \sqrt{t}}\right)$$

$$+ \bar{\Phi}\left(\frac{\log(\frac{y}{s}) - \mu t}{\sigma \sqrt{t}}\right)$$

3.9. Based on the result from 3.8.

$$\frac{y}{s} = \frac{1.2 S'(0)}{S'(0)} = 1.2, \mu = 0.1, \sigma = 0.3$$

$$P(\max_{0 \leq v \leq 1} S'(v) \geq 1.2 S'(0))$$

$$= e^{2 \cdot \log(1.2) \cdot (0.1) / (0.3)^2} \bar{\Phi}\left(\frac{\log(1.2) + (0.1)(1)}{0.3 \sqrt{1}}\right)$$

$$+ \bar{\Phi}\left(\frac{\log(1.2) - (0.1)(1)}{0.3 \sqrt{1}}\right)$$

$$\approx e^{0.4052} \bar{\Phi}(0.9411) + \bar{\Phi}(0.2744)$$

$$= e^{0.4052} (1 - \Phi(0.9411)) + (1 - \Phi(0.2744))$$

$$\approx 0.6518$$

$$\Rightarrow P(\max_{0 \leq v \leq 1} S'(v) < 1.2 S'(0)) = 1 - 0.6518$$

$$= 0.3482$$